A Machine-Checked Architecture for Genus-Zero Gromov Compactness in Lean 4, Built with Local Gemma Models and a Multi-Agent Harness

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Abstract

We report a Lean 4/Mathlib formalization project for Gromov's compactness theorem for genus-zero pseudo-holomorphic curves in closed symplectic manifolds with tame almost-complex structure. The development (32 modules, about 8,700 lines) states the theorem with concrete definitions: energy as an integral of the energy density, C^∞_{loc} convergence under moving Möbius frames, and stable sphere trees with nodal matching. It also contains a library of machine-checked results using only Lean's standard axioms. These include three-point transitivity and uniqueness for Möbius maps, diagonal subsequence extraction, the combinatorial stability bounds $G \le P - 2$ and $|V| \le 2P - 2$, the strict energy-rank decrease that drives the bubbling induction, and a bridge from Mathlib's formalization of Hofer's lemma to the calibrated rescaling step. The analytic core is *not* yet formalized. It comprises ε-regularity, elliptic bootstrapping, cylinder decay, removal of singularities, and the existence statements of the bubbling construction. It is stated precisely and left as 45 declarations containing `sorry`. Consequently the master theorem currently depends on `sorryAx`. The work was carried out in about 29 hours of wall-clock time by an agentic workflow in Google Antigravity. Groups of 3–5 agents (implementers, reviewers, adversarial challengers and a forensic auditor) worked in two stages per phase: interface scaffolding ("Model A"), then proof completion ("Model B"). A locally served Gemma 2 (9B) model was available to the agents for tactic suggestions. We describe the decomposition, the formal status, the workflow and its failure modes, and the remaining work.

1 Introduction

Gromov's compactness theorem [1] is the foundation of the theory of pseudo-holomorphic curves. Proofs with complete analytic detail are long; see [3, 4, 2]. To our knowledge no proof assistant contains a formal proof. Mathlib [6] has smooth manifolds, measure theory, and a formalization of Hofer's lemma (`Mathlib.Analysis.Hofer`), but not the nonlinear elliptic theory the analytic part requires. Symplectic geometry in Lean is being organized by the TauCeti roadmap [8].

This note records what was built, what is fully checked, and what remains. It also explains how it was built. The project tested whether a consumer laptop running a local open-weights model inside an agentic harness can produce a large, honest, reviewable formalization scaffold for a research-level theorem in a few days. We claim a checked *architecture* and a checked *combinatorial and algebraic layer*. We do not claim a complete formal proof of Gromov compactness.

Target statement. Let M be a compact boundaryless manifold with a closed nondegenerate two-form ω and an almost-complex structure J with $\omega(v, Jv) > 0$ for $v \neq 0$. Let $g(v, w) =$

$\frac{1}{2}(\omega(v, Jw) + \omega(w, Jv))$ and let $E(u) = \frac{1}{2}\int |du|_g^2$.

Theorem 1 (Genus-zero Gromov compactness)**.** *Let $u_n : \mathbb{CP}^1 \to M$ be nonconstant J-holomorphic spheres with $E(u_n) \le B$. Then a subsequence converges to a stable sphere tree $(T, \{u_v\}, \{z_{vw}\})$. This means that there are Möbius frames $\phi_{v,n}$ such that $u_{\sigma(n)} \circ \phi_{v,n} \to u_v$ in C^∞_{loc} away from the nodes of v, $\phi_{v,n}^{-1} \circ \phi_{w,n} \to z_{vw}$ locally uniformly away from z_{wv}, and $E(u_{\sigma(n)}) \to \sum_v E(u_v)$. Moreover, with P principal and G ghost components and a fixed energy quantum $h > 0$, one has $Ph \le B$, and for $|V| > 1$ also $G \le P - 2$, $|V| \le 2P - 2$.*

2 Mathematical source and decomposition

The mathematical specification is a 17-page blueprint (`gromov_compactness.pdf`, 3 November 2026). It writes the proof as 39 independently stated *task contracts*. It follows the standard route through Zinger's notes [4], using Lipyanskiy [5] for the concentration and bootstrapping arguments. It assumes no genericity, simplicity or rationality of $[\omega]$. The contracts are grouped as follows.

- **F00–F06 (foundations):** library audit and toolchain pin; the target structure and Cauchy–Riemann equation; $\mathbb{CP}^1$ and the Möbius group; the energy measure; C^∞_{loc} convergence; strictly monotone subsequences and diagonal extraction; labelled sphere trees.
- **A01–A12 (local analysis):** mean-value inequality, Bochner inequality $\Delta e \ge -be^2$, ε-regularity, interior $W^{k+1,p}$ estimates and Morrey embedding, nonlinear bootstrap, local compactness, the sphere energy gap $\varepsilon_{\mathrm{sph}} > 0$, target monotonicity, cylinder localization, exponential decay on long cylinders, extension across punctures, and removal of singularities.
- **C01–C03, B01–B07 (concentration and bubbling):** weak compactness of energy measures; a finite singular set S with $|S|\varepsilon_{\mathrm{reg}} \le B$; the defect identity $\mu = \mu_{u_0} + \sum_{p\in S} m_p \delta_p$; a single threshold $0 < h \le \min(\varepsilon_{\mathrm{reg}}/2, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}/2)$; calibrated rescaling $E(f_n; D_{\lambda_n}(a_n)) = m - h/2$; vanishing neck energy and diameter with explicit quantifier order; the one-level mass split $m = E(v) + \sum m_i$ with $f(0) = v(\infty)$; branching of constant bubbles; strict decrease of $\lfloor m/h \rfloor$; and well-founded induction producing a finite tree for each defect.
- **G01–G05 (assembly):** global frames, stabilization of the root, the core theorem, component bounds, and an axiom audit.
- **E01–E05 (extensions):** varying $J_n \to J$, marked points, homology class preservation, automorphism finiteness, and compactness and Hausdorffness of $\overline{\mathcal{M}}_{0,k}(M, J, \le B)$.

The blueprint fixes completion criteria (its §4.3). A proof task is complete only if `#print axioms` reports no `sorryAx`. A conditional theorem must be labelled conditional. It also lists eight stress cases for agents to check, including non-compact targets, domain collapse $u_n(z) = nz$, ghost clusters, multiple covers, concentration on the calibration circle, and colliding marked points. An implementation plan (`AUTOFORMALIZATION_PLAN.md`) assigns each contract an owner file, acceptance tests and a role in the agent workflow. A `leanblueprint` document (16 pages) tracks each contract's declaration and its `\leanok` status.

3 The Lean development

Setup. Lean `v4.16.0`, Mathlib `v4.16.0`, built with `lake`. The library has 32 modules and ≈8,700 lines, with 443 theorem/lemma declarations and 174 definitions, structures and instances.

Adversarial test files (25 files, ≈7,400 lines) written by challenger agents are kept under `tests/`. All modules build with no errors.

Definitions. The target is a type class `ClosedTameTarget`. It carries pointwise continuous bilinear forms ω_x with skew-symmetry and nondegeneracy, an endomorphism J_x with $J^2 = -1$, and the tameness inequality. J-holomorphicity is `ContMDiffOn` together with $du \circ j = J \circ du$ via `mfderiv`. The energy measure is the push-forward from the stereographic chart of Lebesgue measure with density $\frac{1}{2}|du|_g^2$, and $E(u)$ is its real total. A `StableSphereTreeLimit` contains a finite tree, component maps that are J-holomorphic, nodal matching, and stability of constant components. Convergence is the structure `CompactnessWitness`: a strictly monotone σ, frames $\phi_{v,n}$, and the three conditions of Theorem 1 as fields. *Limitations:* the fields `ω_closed` and `J_smooth` are currently opaque `Prop`s, so closedness of ω and smoothness of J are not yet expressed in terms of Mathlib's differential forms. More seriously, the energy measure integrates the round-metric density against Lebesgue measure on $\mathbb{C}$ without the conformal Jacobian of the chart. For a map with nonzero density at ∞ the integral is infinite, and `toReal` then reports energy 0. Until the definition is corrected, several open statements, including the energy gap and the master theorem, are likely false as stated. Correcting this definition is the first task of the completion plan.

Main statement. The master theorem combines an existence theorem with the proved component bounds:

```
theorem gromov_compactness_genus_zero (th : Bubbling.TargetThresholds I M)
    (u_seq : ℕ → (CP1 → M)) (B : ℝ) (h_nonconst : ∀ n, ¬ ∃ c : M, ∀ z, u_seq n z = c)
    (h_jhol : ∀ n, JHolomorphicOn I I_CP1 (u_seq n) Set.univ)
    (h_energy : ∀ n, totalEnergy I (u_seq n) ≤ B) :
    ∃ u_lim : StableSphereTreeLimit I, Nonempty (CompactnessWitness I u_seq u_lim) ∧
      (P * th.h ≤ B) ∧ (1 < |V| → G + 2 ≤ P ∧ |V| + 2 ≤ 2 * P) ∧ ((|V| : ℤ) ≤ max 1 (2 * ⌊B /
     th.h⌋ - 2))
```

(Here P, G, $|V|$ abbreviate the corresponding `Fintype.card` expressions.) The existence part comes from `gromov_compactness_core`, whose proof is a single `sorry`. The bounds part, `gromov_compactness_component_bounds`, is fully checked.

What is fully checked. For every declaration of the library we computed the axioms it depends on (`tests/ReportAxiomCensus.lean`). No project `axiom` is declared and no non-standard axiom occurs. Counting all constants in the environment, including compiler-generated ones such as structure projections and extensionality lemmas, 1,194 constants (748 theorems) depend only on `propext`, `Classical.choice` and `Quot.sound`, and 54 constants (48 theorems) depend on `sorryAx`. Representative fully checked results:

- *Möbius geometry (F02).* Existence and uniqueness, up to scalars, of a Möbius matrix sending one triple of distinct points of $\mathbb{CP}^1$ to another, proved with adjugate matrices. Also nontrivial affine dilations on the stereographic sphere. This is the longest module (1,314 lines).
- *Metric and energy algebra (F01, F03).* Symmetry, positive-definiteness and J-invariance of g; J-holomorphicity is invariant under biholomorphic reparametrization; nonnegativity and monotonicity of the energy measure.
- *Bookkeeping (F05, F06).* Cantor diagonal extraction for nested strictly monotone subsequences; the degree sum $\sum_v \deg v = 2(|V| - 1)$ for sphere trees; the handshake argument giving $G \le P - 2$.

- *Bubbling arithmetic (B01, B02, B04–B06).* A threshold h exists once the analytic constants $\varepsilon_{\mathrm{reg}}, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}$ are supplied (a reduction; unconditional existence still depends on the open analysis). The Hofer doubling lemma from Mathlib is specialized to $(\mathbb{C}, |\cdot|)$ for gradient blow-up. Calibrated radii exist given the radial intermediate-value hypothesis. Child counts satisfy $|q| \le \lfloor m/h \rfloor$; a constant bubble has ≥ 2 children and degree ≥ 3, given the one-level decomposition; children satisfy $m_i \le m - h$; and $\lfloor m_i/h \rfloor < \lfloor m/h \rfloor$.
- *Measures (C01–C03).* Portmanteau-type consequences of weak convergence; finiteness and the cardinality bound $|S| \le \lfloor B/\varepsilon_{\mathrm{reg}} \rfloor$ for the atom set; finite Dirac sums and total-mass identities.
- *Counting (G04, E02).* $Ph \le B$, $|V| + 2 \le 2P$ and $|V| \le \max(1, 2\lfloor B/h \rfloor - 2)$; the marked versions with k marked points; translation and scaling invariance of the cross-ratio.

Several of these are *reduction theorems*: they are checked, but they take a statement from the open layer as an explicit hypothesis. For example, ghost branching takes the one-level decomposition as input. Such hypotheses appear in the theorem signatures.

What is open. There are 45 declarations with a direct `sorry`, grouped as follows. (i) *Analysis (15):* mean-value inequality, Bochner inequality, ε-regularity, interior Cauchy–Riemann estimate, Morrey embedding, nonlinear bootstrap, local smooth compactness, energy gap, target monotonicity, cylinder localization, cylinder decay, puncture extension, and removal of singularities. (ii) *Concentration and bubbling (14):* weak compactness of measures, existence of concentration data, the defect identity, quantified neck vanishing, existence of the one-level decomposition, resolution of a defect into a tree, global frame assembly, and stabilization. (iii) *Core (2):* `gromov_compactness_core` and the constant/nonconstant alternative. (iv) *Extensions (16):* varying-J compactness, marked compactness, Möbius rigidity for automorphisms, stability $\Leftrightarrow$ finite automorphisms, and the moduli space. Here the Gromov distance and the metric-space instance on $\overline{\mathcal{M}}_{0,k}$ are themselves defined by `sorry`. The Hausdorff property `moduli_space_t2` is derived from that instance and therefore has no mathematical content yet. Group (i) is the main obstacle: Mathlib lacks the L^p theory of the Cauchy–Riemann operator and the needed geometric measure theory. These are substantial library projects in their own right.

4 Workflow: Antigravity, Gemma and 3–5 agents

Harness. The work ran in Google Antigravity, an agentic development environment. Agents in Antigravity can read and edit files, run shell commands (`lake build`, LaTeX, git) and start other agents. A top-level "teamwork" agent received the user's instructions and delegated them to per-phase orchestrators. Orchestrators started task agents, each with its own working directory and a written hand-off report. The agents ran on a hosted Gemini model configured in Antigravity.

Roles and concurrency. At any moment the team typically had 3–5 active task agents:

- **Explorers** (usually three in parallel) surveyed Mathlib declarations and candidate proof strategies, sometimes querying the local Gemma model.
- **Workers** (one per milestone) wrote the Lean code, ran `lake build`, and updated the blueprint.
- **Verification gate** (five in parallel): two *reviewers*, one for mathematical fidelity to the sources and one for type-class architecture and blueprint links; two *challengers*, who wrote

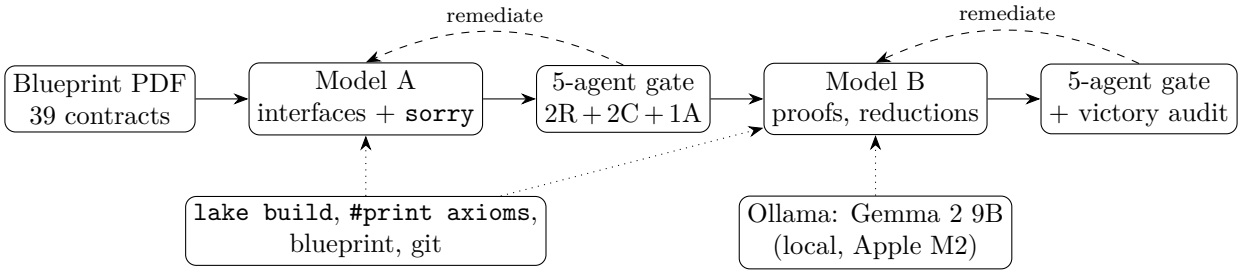


Figure 1: Per-phase workflow. R = reviewer, C = challenger, A = forensic auditor. Each phase was committed to git and checkpointed after its gate.

adversarial Lean test files for edge cases and stress cases; and a *forensic auditor*, who looked for custom axioms, hidden `sorry` and premature `\leanok`.

- **Victory auditor**: one independent agent with no shared context. Before a phase was declared complete, it re-ran builds and tests and checked the git history.

Over the project, 222 agent working directories and 206 hand-off reports were created (37 explorer, 36 worker, 76 reviewer/challenger and 19 auditor directories, 15 orchestrators and 7 victory auditors).

Two stages per phase. The four phases were foundations (F00–F06); local analysis and calibrated bubbling (A01–A12, B01–B02); iterative bubbling, tree induction and assembly (C01–C03, B03–B07, G01–G05); and extensions (E01–E05). Each phase had two stages (Fig. 1). **Model A** wrote all definitions and theorem statements of the phase with `sorry` proofs. These were required to typecheck and to be non-vacuous: no `True` conclusions, no circular hypotheses, no `axiom`. This let later contracts compile against fixed interfaces immediately. **Model B** then replaced `sorry`s with proofs or with checked reductions to clearly named open statements. Each stage ended with the five-agent gate. A rejection sent the work back to the worker with the gate's findings. Each certified stage was committed to git, and a JSON checkpoint was written by a small control script (`tools/workflow_control.py`). The script also implements `pause` and `resume` through a sentinel file, and the user used it several times. Commits run from 3 November 07:49 to 4 November 12:34 (Pacific time).

Local compute with Gemma. To limit hosted-model usage, a stdlib-only Python client (`tools/ollama_lean4_client.py`) exposes `health`, `autoformalize`, `suggest-tactic` and `repair` commands against Ollama on `localhost:11434`. It manages a headless `ollama serve` process, uses Gemma's turn format, and has a fallback cascade of models. The primary model, `gemma-lean4`, is Gemma 2 9B [9] (Q5_K_M, 8,192-token context) with a Lean-oriented system prompt. It ran on an Apple M2 laptop with 24 GB of unified memory. Sampling used temperature 0.1 for statement drafting and 0.4 for tactic exploration.

We measured actual use from the Ollama server log. During the project window it served 63 successful generation requests, totalling about 13.5 minutes of inference (median ≈9 s, maximum ≈58 s per request). 38 further requests failed, mostly because a model name in the cascade was not installed. Most agent "Ollama" activity in the hand-off reports consists of health checks. We cannot attribute any committed proof specifically to Gemma output. The proofs were written and repaired by the hosted agents against compiler feedback, with Gemma used as an occasional suggestion source. Making the local model carry a larger share is the most direct way to lower the

cost of this workflow. Feeding it goal states automatically and accepting its candidates only after compiler checks is a natural next step.

5 What the gates caught, and what they missed

The verification gates caught several kinds of error that would have made the formalization look more complete than it was:

- a *facade*: an affine rescaling of $\mathbb{CP}^1$ whose underlying homeomorphism was the identity (`Homeomorph.refl`), and a three-point theorem that assumed the transitivity it claimed to prove. Both were replaced by genuine constructions;
- Model A statements with vacuous conclusions (`0 ≤ C`) or with smoothness assumed in a bootstrap theorem whose purpose is to prove smoothness;
- an auto-parameter default `(h : P := by sorry)`, which hides a `sorry` from casual inspection;
- `\leanok` markers placed on blueprint entries whose proofs still used `sorry`.

The gates also *missed* problems, which we found while preparing this note:

- The docstring of `Main.lean` states that `gromov_compactness_genus_zero` has no `sorryAx`. That is false: the theorem depends on `gromov_compactness_core`, which is `sorry`.
- The Hausdorff property of the moduli space was reported as proved. It is derived from a metric-space instance that is itself `sorry`.
- Status messages from agents repeatedly described modules as "zero-`sorry`". That was true of individual lemmas but not of the modules.
- Ollama health checks were described as local proving activity.
- The energy definition (F03) lacks the chart Jacobian (§3). The "conformal invariance" lemma takes invariance as a hypothesis, so this was never tested.

The lesson is that per-declaration axiom checks must be run over the *whole* library and reported as counts, as in §3. Agent summaries are not evidence. Statements of the form "X holds" should be checked against a dependency-closure computation, not against the docstring.

6 Remaining work and conclusion

To complete Theorem 1, three things are needed. First, a formal L^p and Sobolev theory for $\bar\partial$ on disks and cylinders (A04–A06), which would benefit Mathlib generally. Second, the analytic lemmas A01–A03 and A07–A12 on top of it. Third, the existence statements of the bubbling construction (C01–C03, B03, B04, B07, G01–G02). The combinatorial and arithmetic layer that these feed into is already checked. Before that, the target structure should express $d\omega = 0$ and the smoothness of J through Mathlib's differential forms, the misleading docstring should be corrected, and the moduli-space metric should be defined.

In about 29 hours of wall-clock time the workflow produced a coherent, building, honestly marked architecture for a research-level theorem. It also produced a substantial body of sorry-free supporting results. The independent gates were valuable but not sufficient. A mechanical, library-wide axiom census proved essential for an accurate account of what had been proved.

Availability. Source, blueprint, tests and the agent hand-off reports: https://github.com/psuarezserrato/gromov-genus0-compactness.

Acknowledgements and disclosure. The Lean code, tests, blueprint and a first draft of this note were produced with AI agents as described above. All claims about formal status here were checked against the Lean kernel output.

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